

# RCIM Model-Bank Reproduction Forward Open-Cell Repair Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

## Overview

- campaign name: track1\_forward\_open\_cell\_repair\_campaign\_2026-04-27\_13\_08\_10
- planning report: doc/reports/campaign\_plans/track\_1/exact\_paper/2026-04-27-13-00-21\_track1\_forward\_open\_cell\_repair\_campaign\_plan\_report.md
- queue size: 300
- execution window: 2026-04-27T16:09:50.0377746+02:00 to 2026-04-27T22:01:36+02:00
- campaign output directory: output/training\_campaigns/track1/exact\_paper/forward\_open\_cell\_repair/track1\_forward\_open\_cell\_repair\_campaign\_2026-04-27\_13\_08\_10
- validation root: output/validation\_checks/paper\_reimplementation\_rcim\_original\_dataset\_exact\_model\_bank\_forward\_open\_cell\_repair
- report timestamp: 2026-04-27-22-32-10

## Executive Summary

- The forward-only repair wave completed the full 300/300 queue successfully.
- Targeted family-target pairs: 30 .
- Promoted pair winners: 11 .
- Retained baseline pair winners: 19 .
- Forward non-green cells moved from 11 to 11 .

## Family Best Retry Outcome

| Family | BestRun                                                   | Scope     | Harmonic | ClosureScore | Met | Near | Open |
|--------|-----------------------------------------------------------|-----------|----------|--------------|-----|------|------|
| SVM    | track1_forward_svm_ampl_h40_open_cell_repair_attempt_07   | amplitude | 40       | 1.000        | 2   | 0    | 0    |
| ERT    | track1_forward_ert_ampl_h156_open_cell_repair_attempt_02  | amplitude | 156      | 0.750        | 1   | 1    | 0    |
| GBM    | track1_forward_gbm_ampl_h3_open_cell_repair_attempt_07    | amplitude | 3        | 1.000        | 2   | 0    | 0    |
| HGBM   | track1_forward_hgbm_ampl_h0_open_cell_repair_attempt_06   | amplitude | 0        | 1.000        | 2   | 0    | 0    |
| XGBM   | track1_forward_xgbm_ampl_h240_open_cell_repair_attempt_06 | amplitude | 240      | 0.500        | 0   | 2    | 0    |
| LGBM   | track1_forward_lgbm_ampl_h81_open_cell_repair_attempt_08  | amplitude | 81       | 1.000        | 2   | 0    | 0    |

## Campaign Best Run

- run: track1\_forward\_lgbm\_ampl\_h81\_open\_cell\_repair\_attempt\_08
- family: LGBM
- scope: amplitude
- harmonic: 81
- closure score: 1.000
- met / near / open: 2 / 0 / 0

## Forward Benchmark Delta

| Surface                        | Before        | After         | Delta   |
|--------------------------------|---------------|---------------|---------|
| Table 2 forward amplitude MAE  | 94G / 3Y / 3R | 94G / 3Y / 3R | 0Y / 0R |
| Table 3 forward amplitude RMSE | 96G / 3Y / 1R | 96G / 3Y / 1R | 0Y / 0R |
| Table 4 forward phase MAE      | 89G / 1Y / 0R | 89G / 1Y / 0R | 0Y / 0R |
| Table 5 forward phase RMSE     | 90G / 0Y / 0R | 90G / 0Y / 0R | 0Y / 0R |

## Canonical Forward Status After Closeout

- Table 2 : 3 yellow, 3 red
- Table 3 : 3 yellow, 1 red
- Table 4 : 1 yellow, 0 red
- Table 5 : 0 yellow, 0 red

## Reference Archive Refresh

| Direction | Family | Archived Targets | Unique Source Runs | Unique Source Configs | Archive Root                                                     |
|-----------|--------|------------------|--------------------|-----------------------|------------------------------------------------------------------|
| forward   | SVM    | 19               | 11                 | 11                    | models/paper_reference/rcim_track1/forward/svm_reference_models  |
| forward   | MLP    | 19               | 13                 | 13                    | models/paper_reference/rcim_track1/forward/mlp_reference_models  |
| forward   | RF     | 19               | 12                 | 12                    | models/paper_reference/rcim_track1/forward/rf_reference_models   |
| forward   | DT     | 19               | 12                 | 12                    | models/paper_reference/rcim_track1/forward/dt_reference_models   |
| forward   | ET     | 19               | 12                 | 12                    | models/paper_reference/rcim_track1/forward/et_reference_models   |
| forward   | ERT    | 19               | 10                 | 10                    | models/paper_reference/rcim_track1/forward/ert_reference_models  |
| forward   | GBM    | 19               | 11                 | 11                    | models/paper_reference/rcim_track1/forward/gbm_reference_models  |
| forward   | HGBM   | 19               | 11                 | 11                    | models/paper_reference/rcim_track1/forward/hgbm_reference_models |
| forward   | XGBM   | 19               | 11                 | 11                    | models/paper_reference/rcim_track1/forward/xgbm_reference_models |
| forward   | LGBM   | 19               | 17                 | 17                    | models/paper_reference/rcim_track1/forward/lgbm_reference_models |

## Linked Artifacts

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- benchmark report: `doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md`
- training results master summary: `doc/reports/analysis/Training Results Master Summary.md`
- track1 reference root: `models/paper_reference/rcim_track1/forward/`
- campaign leaderboard:  
`output/training_campaigns/track1/exact_paper/forward_open_cell_repair/track1_forward_open_cell_repair_campaign_2026-04-27_13_08_10/campaign_leaderboard.yaml`
- campaign best run YAML:  
`output/training_campaigns/track1/exact_paper/forward_open_cell_repair/track1_forward_open_cell_repair_campaign_2026-04-27_13_08_10/campaign_best_run.yaml`
- campaign best run Markdown:  
`output/training_campaigns/track1/exact_paper/forward_open_cell_repair/track1_forward_open_cell_repair_campaign_2026-04-27_13_08_10/campaign_best_run.md`